

Marha Midhatiey

Specialized expertise in LLM, Prompt Engineering, and developing deep learning solutions, Computer Vision, Sentiment Analysis and Voice Synthesis projects. Website: <https://mareuu.github.io/>

Buk-Gu, Gwangju
South Korea, 61005
(+82) 010-9518-7335 (Main)
(+60) 019-357-1395
marhamidhatiey@gmail.com

EXPERIENCE

Gwangju Institute of Science and Technology, Gwangju, South Korea—*Research Assistant (Data Science Lab)*

2024 - 2026

- Researched Hybrid AI for visual reasoning, focusing on symbolic and neural methods to improve AI interpretability and cognitive ability.
- Optimized large-scale model training on HPC clusters, significantly enhancing computational efficiency for complex benchmark reasoning tasks.
- Led experimental design and implementation of deep learning models, ensuring robust evaluation of hybrid reasoning architectures.

3F Resources Sdn.Bhd., Kuala Lumpur, Malaysia—*System Analyst & Database Engineer (Internship)*

2023

- Developed and designed new systems and frameworks(UI/UX, UML Databases) while actively participating in the maintenance and upgrading of existing systems.
- Functioned as a Database Engineer and System Analyst, contributing to critical government projects for the Ministry of Education.

Rifhan Teknologi Sdn.Bhd., Kuala Lumpur, Malaysia—*System Administrator & Maintenance (Internship)*

2020

Administered and maintained operational systems for three major government agencies, ensuring system stability (at Ministry of Home Affairs, Royal Malaysia Police Headquarters, and the Ministry of Finance).

EDUCATION

Gwangju Institute of Science and Technology, Gwangju, South Korea—*Master's in Artificial Intelligence*

2024 - Feb 2026

SKILLS

Generative AI : LLM

Fine-Tuning (QLoRA, SFT, LoRA), Prompt Engineering

AI/Deep Learning: Enhancing LLM Reasoning, PyTorch, Transfer Learning, CNNs, MelBandRoformer, Kafka Integration

Programming: Python, R, Java, C++, PHP, Visual Basic

Database/BI: Oracle Database, MySQL, Oracle SQL Developer, IBM SPSS, Weka, Microsoft Power BI

Analytics/Tools: Big Data Analytics, RapidMiner, Web Scraping Tools, Gradio, Git

AWARDS

Best Paper Award for publication in Journal of Advanced Research in Applied Mechanics (ISETech 2023)

Vice Chancellor Awards in Bachelor's degree

Dean's Award every semester in Bachelors's degree and Associate years

LANGUAGES

English(Native), Malay(Native), Korean(Conversational)

**Universiti Teknologi MARA (UiTM), Selangor, Malaysia—
Bachelor of Information Systems (Hons.) Intelligent Systems
Engineering**

2021 - 2023

GPA: 3.79/4.00 ; Major in Big Data & Artificial Intelligence

**Universiti Teknologi MARA (UiTM), Melaka, Malaysia—
Diploma in Computer Science**

2018 - 2021

GPA: 3.83/4.00

PROJECTS & PUBLICATIONS

**Tracing and Correcting Programs : Critic-Guided Synthesis
for Visual Reasoning (LLM Focus) — AAAI 2026 Bridge
Program(Accepted & Presented), Master's Thesis works**

Research focused on LLM Interpretability and Reasoning through advanced Prompt Engineering and adaptive signal techniques for LLM.

Experienced in fine-tuning small LLM models using techniques like QLoRA, Supervised Fine-Tuning (SFT) and LoRA to upskill model capabilities and enhance output quality.

Publication Note: A part of this academic thesis is currently under review at the Data Mining and Knowledge Discovery.

**Driver Drowsiness Detection using Transfer Learning and
Computer Vision (Best Paper Award) — Publications**

Designed a computer vision system to detect driver drowsiness by monitoring eye closure using **Transfer Learning**. Engineered the system to analyze video frame-by-frame via webcam feed, triggering an alert if eyes were closed for more than 10 consecutive frames.

Publication: Journal Of Advanced Research in Applied Mechanics
(ISETech 2023)

How You Sound Like? — Voice Synthesis System

Developed an end-to-end system to detect user voices and match them with target singers based on tone and pronunciation.

Utilized advanced deep learning technologies including **MelBandRoformer** and **PyTorch**, integrating **Gradio** for the UI and **Kafka** for robust data handling.

Sentiment Analysis On Malaysia Airlines

Conducted comprehensive sentiment analysis, utilizing RapidMiner on data retrieved from multiple Malaysia Airlines websites for analytical purpose.